

## Demonstration of Proposed Features

|                      |               |
|----------------------|---------------|
| <b>Date</b>          | 15 March 2026 |
| <b>Team ID</b>       | XXXXXX        |
| <b>Project Name</b>  | SmartLender   |
| <b>Maximum Marks</b> | 1 Mark        |

### Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

| S.No | Feature Name                      | Description                                                                                                                                                                              | Status (Implemented / Partial / Pending) | Demonstrated (Yes / No) | Remarks                                                            |
|------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------------------------|--------------------------------------------------------------------|
| 1    | User Registration & Login         | Secure user authentication allowing applicants to register and access the platform                                                                                                       | Implemented                              | Yes                     | Successfully demonstrated during the live web app walkthrough.     |
| 2    | Automated Data Preprocessing      | Pipeline that handles missing values, encodes categories, applies StandardScaler, and balances classes using SMOTE.                                                                      | Implemented                              | Yes                     | Demonstrated the background execution via                          |
| 3    | ML Model Training & Serialization | Automated training of multiple algorithms (Decision Tree, RF, KNN, XGBoost) and serialization of the best model                                                                          | Implemented                              | Yes                     | Model successfully loaded into the application environment.        |
| 4    | Web Application Interface         | Responsive Flask frontend allowing users to input their financial and personal details for evaluation                                                                                    | Implemented                              | Yes                     | Demonstrated smooth UI interaction using custom CSS and JS.        |
| 5    | Real-Time Prediction Inference    | The /submit endpoint processes data in real-time to output a final "Loan Approved <input checked="" type="checkbox"/> " or "Loan Rejected <input checked="" type="checkbox"/> " decision | Implemented                              | Yes                     | Handled live user inputs perfectly during the demo phase           |
| 6    | User Dashboard & Analytics        | Dedicated views for users to see analytics and manage their prediction history                                                                                                           | Implemented                              | Yes                     | Provided excellent visualizations of the loan prediction pipeline. |

**Feature Implementation Summary:**

|                                 |      |
|---------------------------------|------|
| Total Features Proposed         | 6    |
| Total Features Implemented      | 6    |
| Total Features Demonstrated     | 6    |
| Overall Implementation Rate (%) | 100% |